Executive Summary – Iris Classification

Project Objective

The objective of this project was to build a machine learning classification model capable of predicting Iris flower species using sepal length, sepal width, petal length, and petal width measurements.

Dataset

The Iris dataset contains 150 observations across three flower species:

- Iris Setosa
- Iris Versicolor
- Iris Virginica

Exploratory Data Analysis

EDA showed that petal length and petal width are the most discriminative features. Pair plots and scatter plots demonstrated clear class separability, especially for Iris Setosa.

Models Evaluated

- k-Nearest Neighbors (k-NN)
- Logistic Regression
- Decision Tree Classifier

Evaluation Metrics

Models were evaluated using:

- Accuracy
- Precision
- Recall
- Confusion Matrix

Results

All models achieved high classification performance, with the best-performing model selected based on test accuracy. The confusion matrix indicated minimal classification errors.

Conclusion

The project successfully demonstrated the effectiveness of machine learning algorithms for multiclass classification. Petal measurements were found to be the strongest predictors of Iris species. The final model was saved and can be used for future predictions.